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SMART WATER SURFACE CLEANING BOAT USING MOBILE APPLICATION

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ABSTRACT

Water plays an essential role in our daily lives and is vital for sustaining life. Many towns and communities rely on rivers and other natural water bodies as major sources of drinking water. Unfortunately, the increasing amount of waste found in these waters is not just an inconvenience it poses serious dangers to the environment, to us, and to our loved ones. Even a single piece of litter thrown on the ground can eventually end up in streams and rivers, contributing to pollution. Therefore, it is necessary to remove this waste efficiently. The Water Cleaning Boat Project aims to clean rivers and lakes by Remote controller, with the goal of reducing pollution while minimizing human effort. In this project, a special boat is designed, built, and tested. It uses a conveyor belt driven by a DC motor to pick up floating garbage like plastic, leaves, and other waste from the surface of the water. The boat's electrical system is controlled by a Motor Driver and ESP32, which can be operated wirelessly through a mobile phone.

Keywords: Water Cleaning Boat, Environmental Protection, ESP32, Waste Collection, Water Pollution Control.

I. INTRODUCTION

As we all are aware about 71 percent of the Earth is covered with water, and it has always been extremely important for human life. From ancient times, people have lived near rivers and lakes because they help us survive and grow. Even today, we depend on water bodies for many needs. However, with increasing industrialization, we have started harming these natural resources. Wastewater often contains materials that do not decompose. These non-biodegradable wastes are dangerous because they can cause climate changes, floods, and pollution. To our environment, it is our responsibility to save and maintain these resources. For many years, people have dumped waste into water bodies, causing heavy pollution. Now it is time to clean up the mess we created. This is why we need a smarter and more eco-friendly system. One such solution is a water cleaning boat. This machine can remove waste from water bodies with very little human effort. It collects floating waste from lakes, rivers, and ponds.

II. METHODOLOGY

The methodology of the water cleaning boat explains the steps used to design, develop, and operate the system that collects floating waste from water bodies.

Components ESP32

- Here ESP32 act as a centre unit of the project. The ESP32 can connect with other components to offer Wi-Fi and Bluetooth features.
- It receives wireless commands from a smartphone app or a remote controller through Bluetooth or Wi-Fi.
- These commands allow the operator to control how the boat moves and which direction it goes.

Motor Driver

Connected to the ESP32 to control the speed and direction of the motors. **DC Motors**

Used for two purposes: boat movement and running the conveyor belt. **Conveyor**

Belt

- The conveyor belt uses the motor's torque to drive a chain mechanism that lifts waste from the water's surface and drops it into a storage bin called the garbage collector.
- PVC pipes are placed underneath the boat to support the air-tube guiding mechanism.

- This setup directs floating debris toward the boat, where the conveyor belt then gathers and moves the waste into the collector. **Lithium Battery**
- A lithium-ion battery is used as the main power source for the water-cleaning bot. The lithium battery provides DC power to gear motors, which drive the propellers for moving the boat forward/backward. The conveyor belt mechanism that collects floating waste. Because lithium batteries give a stable voltage, the motors run smoothly and efficiently.

Boat Movement System

The ESP32 sends signals to the motor driver, which controls the DC motors attached to propellers. These motors help the boat move forward, backward, left and right on the water surface.

Waste Collection process

When the boat moves toward floating waste, the conveyor belt rotates. The belt lifts the waste materials from the water. The collected waste transferred to a storage container inside the boat.

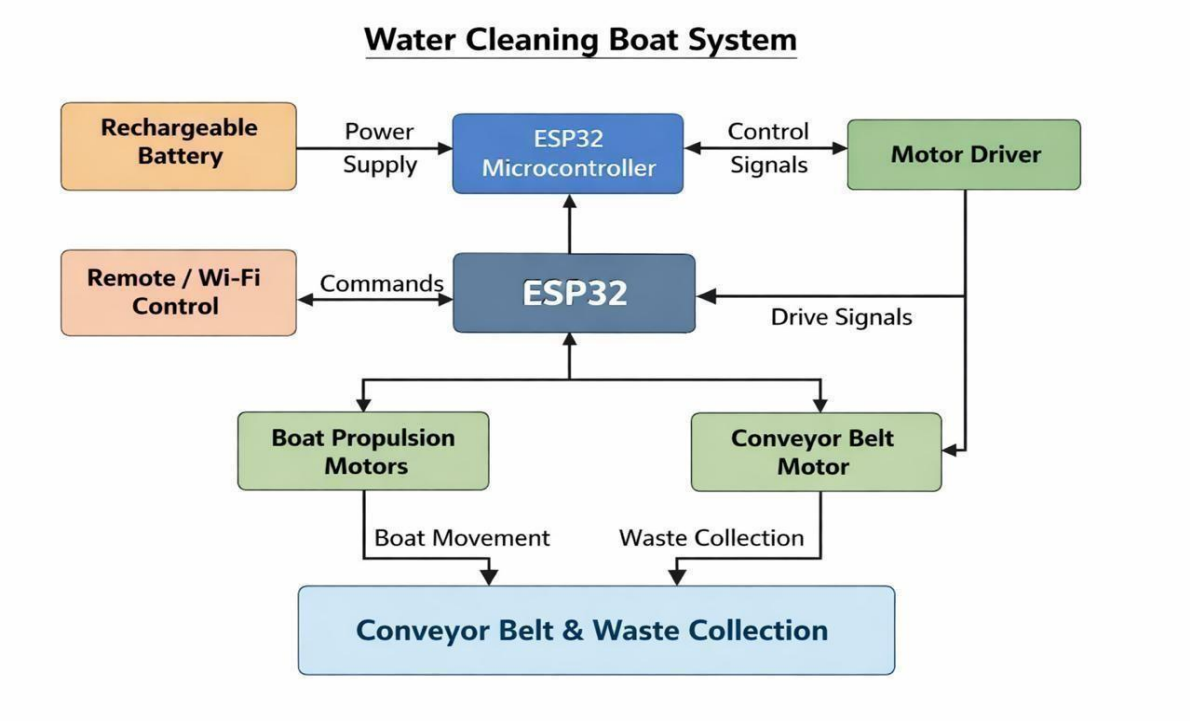


Figure 1: Block Diagram Control

System

The ESP32 processes commands from a mobile app and sends control signals to the motor driver the motors and conveyor belt.

Power Supply

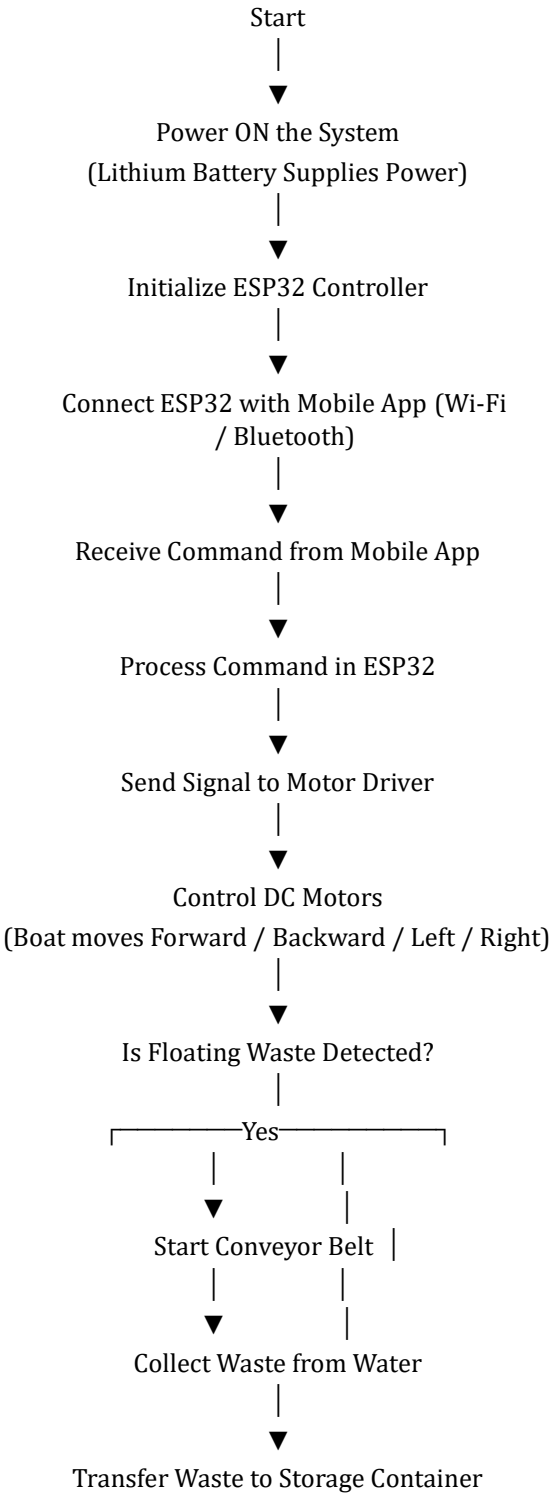
A rechargeable battery supplies power to the ESP32, motor driver and DC motors run the entire system.

III. MODELLING AND ANALYSIS

The water cleaning boat is designed to remove floating waste from water bodies such as lakes, rivers, and ponds. The system mainly consists of an ESP32 microcontroller, motor driver, DC motors, a conveyor belt mechanism, and a lithium battery for power supply. In the modelling of the system, the boat is considered as a floating platform that moves on the water surface using DC motors for propulsion. The ESP32 microcontroller acts as the main control unit that receives commands from a mobile application through wireless communication such as Wi-Fi or

Bluetooth. These commands are processed by the ESP32 and transmitted to the motor driver, which controls the direction and speed of the DC motors to move the boat forward, backward, left, or right.

The waste collection modelled using a conveyor belt system placed at the front of the boat. When the boat moves across the water surface, floating waste such as plastic bottles, leaves, and other debris is guided onto the conveyor belt. The belt is driven by a DC motor, which rotates the rollers and lifts the waste from the water surface into a storage container inside the boat.



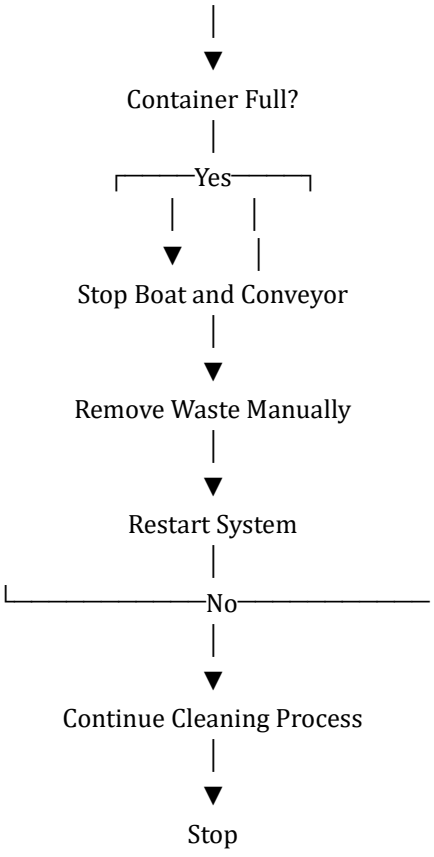


Figure 2: Flow chart

IV. RESULTS AND DISCUSSION

The use of a water cleaning boat based on mobile application has shown a positive results. It effectively removes trash, leaves, and other floating debris from rivers, lakes, and ponds, making the water clean and safe for the aquatic life. It not only reduces pollution but also helps to protect the environment and maintain better water quality. The mobile application allows real-time monitoring, showing areas that have been cleaned and the amount of waste collected, which makes the cleaning process more efficient. Additionally, the automated operation saves time and reduces the need for manual labour. Overall, this technology provides an eco-friendly solution for maintaining clean water bodies and also being the effective step in order to decrease the water pollution.

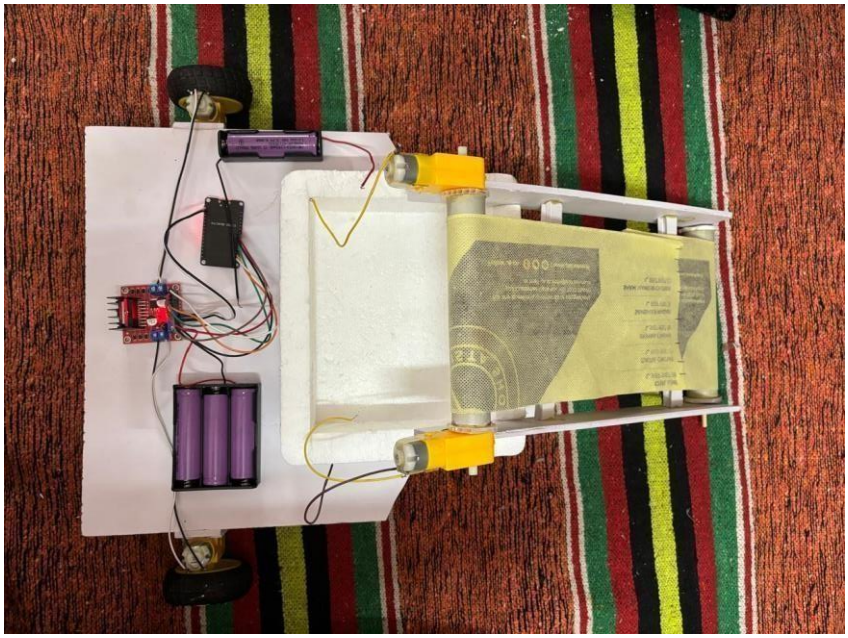


Figure 3: Result of water cleaning boat

Sr No.	Condition	Output
1	Power Supply is ON	ESP32 and Motor driver start functioning
2	ESP32 receives command to move forward	Boat moves forward
3	ESP32 receives command to move backward	Boat moves backward
4	ESP32 receives command to move left	Left motor slows and boat turns left
5	ESP32 receives command to move right	Right motor slows and boat turns right
6	Conveyor belt is activated	Conveyor belt starts rotating
7	Floating waste reaches conveyor belt	Waste is lifted from water

Table 1: performance Result

V. CONCLUSION

Water is one of the most critical naturally occurring resources and is increasingly subjected to contamination from multiple sources. Surface floating debris obstructs sunlight penetration, disrupting photosynthesis and negatively impacting aquatic life, ultimately degrading the water quality. To maintain a healthy aquatic ecosystem, it is essential to remove all surface debris efficiently. In this system, a conveyor mechanism is employed to collect floating waste and deposit it into a storage container. The boat's wireless control system regulates both the conveyor and propulsion motors,

enabling automated debris collection. This integrated approach provides an efficient, cost-effective solution for surface water debris management, enhancing environmental sustainability and ecosystem health.

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